

The Evolution of Shadow Banking System in Emerging Economies: The Role of Entrusted Loans in China's Capital Market

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Abstract

In this paper, we analyze the Chinese economic growth model post-2008 and more notably the effect of 4 Trillion Yuan stimulus in the aftermath of the Global Financial Crisis. We look at the Chinese financial system and the credit creation mechanism in the economy. The data shows that post-2008 there has been a sudden increase in credit outside the formal banking system. We specifically focus on the off-balance sheet Entrusted Loans provided by non-bank entities via the shadow banking system. We explain the entrusted loan mechanism and the rapid increase in Non-bank credit. Our stress tests results for the 15 major banks in China indicate that the banks remain well capitalized in the case of an idiosyncratic shock, notably the SIFI's but can experience contagion through the Interbank funding market via smaller banks.

Keywords: China, Shadow Banking, Banks, Stress Tests, Off-Balance Sheet Investment Vehicles

JEL Classification: G15, G23, G28, G38

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1. Introduction

The importance of a financial system for economic growth traces back its origins to the classical work of [Schumpeter \(1912\)](#) which argues that well-functioning banks spur technological innovation by identifying and funding those entrepreneurs that potentially have the best chances of successfully implementing innovative products and production processes. Henceforth the main aim of a financial system and its regulation in a market economy is to direct the scarce savings of the society, including depreciation plus borrowed savings from abroad (the current account deficit), towards potentially productive intellectual and physical investments. When competition is pervasive and allowed to function, the investments with the highest rates of prospective return and least variance of expected profit, promise to contribute the most to growth in output per hour. It is this gap between the level of output per hour embodied in new capital investment, presumably employing latest technologies, and that of obsolescent, low-productivity facilities that over time engenders net gains in average output per hour, and hence increases in real per capita income and standards of living [Greenspan \(2010\)](#). A well-functioning financial system facilitates the allocation of resources, across space and time, in an uncertain environment ([Merton and Bodie \(1995\)](#), p. 12). While [Rajan and Zingales \(2003\)](#) documents the positive influence a country's financial sector has on the level of growth in its per capita income. Their presumption primarily rests on the central premise that the existence of a well-functioning financial sector is to provide services that allow proper allocation of capital and risk in the economy, and which indeed are an important catalyst for economic growth. Therefore, in light of these arguments, it can be reasonably deduced that a financial system has two prime objectives, to place risks where they are best born and to channel resources to their most productive use.

To ensure unfettered flow of resources, it is quintessential that the lender feels confident of the prospect of an adequate return on its investment on a risk-return metric. [Rajan and Zingales \(2001\)](#) have presented a subtle distinction; they have segregated financial system based on the type of relationship between the lender and the borrower. In the

first case, the system is characterized by the lender's proximity to the borrower, combined with some degree of ownership power granted by the firm. This mechanism essentially allows the financier some monopoly power (by having implicit or explicit ownership of the firm), and thereby allows him to secure a return on his investment. This monopoly power is granted by the presence of some barriers to entry, which are in the form of government regulation, a permissive legal system or lack of transparency owing to the opacity of the system. On the other hand, Anglo-Saxon market-based arrangement relies heavily on the protection of the financier by explicit contracts and transparency. Under this mechanism, legally enforceable contracts help in directing/governing the terms of transactions, and at the same time markets, and associated prices take center place in determining the transactions undertaken, and thereby pricing risk efficiently.

In light of these discussions, this paper contributes to the existing literature three folds, first we study the investment-driven economic growth model and the economic viability of Fixed Asset Investment in China post-2008 crisis and the resultant effect of the 4 Trillion Yuan fiscal stimulus provided by the Chinese authorities to the economy. Secondly, we look at the non-bank credit creation through Entrusted Loans in the Shadow Banking System and lastly to understand the impact of entrusted loans on Banks Balance Sheet for Solvency Risk by using a stress test model. Our interpretation of the Marginal Efficiency of Capital is that a major part of 2008 stimulus went into Fixed Asset Investment and, therefore, elevated real estate prices as we see in the decomposition of Fixed Asset Investment into Gross Fixed Capital Formation. Which created significant over-investment in the economy, and persistent supply side deflation with producer prices declining since 2012. Our results for credit gap further suggests that credit to GDP has been expanding and has been running above trend, since 2012. The credit to GDP measure also captures the effect of alternate financing provided by the non-bank sector in the economy, as we observe that bank loan to GDP hasn't risen considerably, but the overall credit in the economy has expanded quite rapidly. Lastly, the Entrusted loans are off-balance sheet loans and do not require banks to assign risk weights or loan loss provisions. Our stress

tests, results show that bank balance sheets are prone to solvency risk due to a rapid deterioration in banks asset quality, in a case where banks are obliged to move off-balance sheet items on their books and allocate capital for loan loss provisions to maintain regulatory CAR (Capital Adequacy Ratio). Our results further indicate that liquidity risk can be an issue for the banking sector as a whole due to bank's exposure to the inter-bank funding channel. The remainder of the paper develops as follows; Section 2 presents stylized facts and a brief discussion of the China growth model, its financial system, real estate and foreign exchange market. Section 3 focuses on the shadow banking system in China, in particular, Entrusted Loans structure and mechanism. We also discuss the methodology for our Bank Stress Tests, our empirical results. Finally, Section 4 presents the conclusion of our study.

2. China's Economic Growth Model

The growth of Chinese economy has been exponential, GDP over the past ten years has registered an average yearly growth rate of more than ten percent³. It's a phenomenal achievement for the world's largest populated country, and for the millions of people alleviated out of poverty with the kind of reforms that have been instituted by the Chinese leadership. However, caution needs to be exercised as to how these exceptional growth rates are achieved and potential black swan events that may derail this growth story, and if these growth rates can sustain over an extended period. For instance, the data demonstrates in figure 1, that there has been a significant change in the share of household spending and investment in the nominal gross domestic product (GDP). For instance household spending which was more than 55% in 1970 has fallen to less than 35% in 2013, whereas the share of investment, on the other hand, has risen from 25% to 45% for the corresponding period⁴. Japan and the Asian Tiger economies (Malaysia, Indonesia, Thailand & South Korea) had reported similar high investment shares (30-40%) as a

³IMF WEO 2015

⁴The empirical data for this paper is from People's Bank of China (PBOC), National Bureau of Statistics in China & CEIC database.

percentage of GDP⁵ before they experienced their crisis. In China high savings rate has been primarily not only due to low household consumption (in the absence of social safety net) but also due to a smaller share of family income in national income. This smaller percentage can be explained by financial sector weakness which rations firms' access to capital from banks and leads to low wage and investment income share. It demonstrates that a lot of Chinese growth story comes from investments, rather than by private household consumption. Therefore to analyze the efficiency of these investments in generating economic activity, it is quintessential to examine the role of financial market intermediation in China, as it is through this system, savings from household are channelized to entrepreneurs (See Prasad (2004)Ayyagari et al. (2010)Song et al. (2011))

2.1. Chinese Financial System

Prior to 2008 much of the financial market intermediation in China happened through the banking system, with a small proportion attributed to the Shadow Banking system or credit creation by Non-Banking Financial Institutions (NBFI's). At present the financial system is plagued by two pressing issues, first concerning the rapidly increasing levels of debt both household and private businesses, and the increasing complexity and opacity of the financial system, specifically its shadow banking system. The term shadow banking was coined by Paul McCulley of PIMCO⁶ at the Federal Reserve Bank of Kansas City, Jackson Hole Economic Symposium in 2007. He defined shadow banks as *the whole alphabet soup of levered up non-bank investment conduits, vehicles, and structures*, in essence shadow banks are non-deposit taking financial institutions that engage in credit, maturity or liquidity transformation without having access to a lender of last resort (section 3 provides a detailed discussion). Historically, credit in the Chinese economy is often provided by the big five Chinese State Owned Commercial Banks (SOCB's) which account for more

⁵WorldBank Database

⁶Pacific Investment Management Company

than 50% of bank lending in the Chinese economy⁷ and three policy banks ⁸ for financing government directed spending (See [Aziz \(2006\)](#)). Since 2008 much of the economic growth in China is generated by increasing accumulation of debt, and with these increasing debt levels, the Chinese government undertook steps to rein lending by banks in the economy. However at the same time banks created complex and opaque structures/arrangements in order to bypass these regulations and provide credit to the firms.

The Chinese government in November 2008 announced a 4 Trillion Yuan (\$586 Billion) stimulus package (12.5% of 2008 GDP) to balance out effects of the global financial crisis on the Chinese economy. However the stimulus package, instead of being utilized to boost household consumption, ended up in an exponential increase in infrastructure spending. The funds were channelized through banks to state-owned enterprises (SOE's) and local governments financing vehicles (LGFV's) to undertake massive infrastructure projects. One of the unintended consequences of this credit boom has been a dramatic surge in corporate debt to Non-Financial Companies (NFC's) which has quadruple from 2007 till 2014, much of this debt went to Large SOE's to undertake massive infrastructure and development projects to generate economic activity⁹. It, therefore, leads to a broader question of analyzing the efficiency of China's long-term fixed non-residential assets over time. As in the end, it is the effectiveness of these income generating capital assets namely industrial plants, structures, etc. that ascertain the sustainability of any economy's investment led growth trajectory over the long term horizon. To do so, we compute the contribution of Fixed Asset Investment (FAI) to GDP. As we see in [figure 2](#), in 2003 Fixed Asset Investment share in GDP stood 45% and eventually rose to more than 80% by the end of 2014. Much of the credit surge has been used to finance rising land prices, which essentially have no contribution in economic activity, therefore, we intend to see the contribution of fixed asset investment excluding the effect of rising land prices the

⁷The Bank of China (BOC), China Construction Bank (CCB), The Agricultural Bank of China (ABC), Bank of Communications and The Industrial and Commercial Bank of China (ICBC).

⁸Agricultural Development Bank of China (ADBC), China Development Bank (CDB) and Export-Import Bank of China (Exim Bank).

⁹China National Audit Office (2013)

proxy is Gross Fixed Capital Formation (GFCF). Both FAI and GFCF measures investment but FAI includes investment in physical asset and property while GFCF measures investment in new equipment and structures excluding the value of land. The difference between FAI and GFCF captures the effect of an increase in land value. While GDP numbers include GFCF, only a portion of FAI is part of the GDP. As we look into the data until 2003 both the measures moved in lock step and, therefore, the effect of land prices was nonexistential. However post 2003 the two measures have diverged while the GFCF stands around at 45% of GDP in 2014, and the share of FAI has risen to more than 80%.

As one would expect the credit levels have also increased in line with the rapid growth in FAI rather than the more modest growth in GFCF. However, it is just the inclusion of land related transactions in the FAI that serves as a significant distinction between the two measures. The difference between the two measures also suggests that such sales account for an increasingly large share of credit, which is consistent with the fivefold increase in the price of land over the past decade. Incidentally, these land transactions do not contribute to GDP, and much of the recent credit growth is due to such transactions. It also explains much of the decline in the growth impact of credit in GDP accounts. We further extend our analysis and look into the Marginal Efficiency of Capital i.e. the Incremental Capital Output Ratio (ICOR) for the Chinese economy from 1996 till 2014 (we follow [Corsetti et al. \(1999\)](#) methodology). ICOR is computed by measuring the change in capital stock relative to the change in output, which essentially normalizes the investment rate relative to the rate of output growth. It measures the amount of additional unit of capital needed to generate one new unit of economic output. As per our calculations (see figure 3) the Chinese economy currently requires more than five units of capital to produce one additional unit of economic output, as compared to a value of two in 2007. It also shows that the efficiency of allocating capital has decreased in the Chinese economy, but also sheds light on how productively credit have been used in generating economic activity, as eventually it is the ratio of capital assets to actual output that determines

the efficiency of long-term income generating assets. Further if the investment boom is limited to the non-tradable sector which essentially is the case in China the sustainability of these projects rests on the indirect impact of the tradable sector's productivity.

Since January 2011, the Industrial Production data on a year on year (YoY) basis has been exhibiting a downward trend. However on a closer look, we observe higher levels of inventory for the same period. This further suggests that there has been excess production capacity in the Chinese economy vis-a-vis the demand in the economy, as seen from the Producer Price Index (PPI) which measures the change in factory gate prices for the corresponding period. The PPI series follows a quarterly lagged relationship with the inventory series. As we see in figure 4 since April 2012, producer prices have been declining in the Chinese economy reflecting a deflationary trend. Deflation is best to define as a general decline in prices, with emphasis on the word "general". Deflation is in almost all cases a side effect of a collapse in aggregate demand—a drop in spending so severe that producers must cut prices on an ongoing basis to find buyers. An exception to this rule is a case when deflation is caused by a sudden, massive expansion in aggregate supply, arising for example from rapid gains in productivity and broadly declining costs. China is currently experiencing a supply-side deflation due to excess production capacity. Also, supply side deflation will be associated with an economic boom rather than a recession as is the case in aggregate demand side deflation.

To further deepen our understanding of economic activity in China, we look at alternate economic indicators namely Bank Lending, Industrial Electricity Consumption and Rail Freight Volume. Bank lending allows one to foresee credit creation in the economy and the remaining two indicators allow one to gauge the health of the manufacturing sector of the economy. When we look at the YoY change for these indicators from 2010, both electricity consumption and rail freight volume exhibit high seasonality during the month of January and February due to the holiday season for the lunar year. The data in figure 5 shows that, for the most part, the indicators have been growing steadily on

a YoY basis. Since October 2013, we see a demonstrable slowdown in the Rail Freight Volume, while electricity consumption has been rising, which suggests that the factories have been producing goods, but they have not been transported to the port cities for exports or to other cities for domestic consumption. Incidentally, when one looks at the inventory levels for the corresponding period, the data shows a rise in inventory levels. Interestingly since 2015 industrial electricity consumption has barely managed to grow while rail freight volume is contracting and is less than 10% on a YoY basis. On the other hand bank lending which is the total amount of credit generated by the formal banking sector is still rising at more than 15% on a YoY basis. It looks trivial in light of the pro-cyclical nature of private credit growth and GDP growth.

The Export and Import data in figure 6 shows that from 2007 till 2015 China has been running big trade surpluses and maintained a market share of more than 14% in Global Export Market by the end of 2013 [IMF \(2015\)](#). Post 2008 the global economy slowed considerably in the aftermath of the financial crisis and global trade still running below its trend level due to reduced demand from the developed economies [World Bank \(2015\)](#). The Chinese export sector suffered a rapid deceleration in 2008; the effect was more profound, due to low household consumption which did not provide the necessary cushion to absorb excess capacity/slack in the economy. The 2008 stimulus provided the investment boom which absorbed the excess labor supply and significantly reduced the effect of the economic slowdown [Schucher \(2009\)](#). For instance according to [World Bank \(2015\)](#) estimates China imports more than 60% of the total production of Nickel and Iron Ore, the two leading industrial commodities. China is also the world's second-largest consumer of crude oil, however it accounts for a meager 18% of its total energy consumption. Therefore, a low crude oil price shock will boost economic activity by 0.1-0.2 percentage points. Since 2011 Exports and Imports have shown a downward trend specifically imports which have been growing negatively since January 2014. This further corroborates the economic slowdown in the Chinese domestic economy, and if one solely looks at the Trade Balance numbers, they may prove deceptive, as one may observe that even though

the trade surplus has increased, it has been solely due to a decrease in the value of imports.

2.2. Real Estate Industry

Article 10 of the 1982 Constitution of China stipulated that all land in China belongs to its citizens as a whole, it further emphasizes that land of the country belongs to the state or its agricultural collectives namely villages. State-owned enterprises (SOE's) or other organizations, which cannot own land themselves, may use it with permission from the state. As the Chinese economy prospered and subsequently with increasing standards of living the demand for increased usage of land grew rapidly. Therefore, in 1988, The People's Republic of China made an amendment to the constitution, under this bill private investors can obtain land use rights from the government and can then retransfer them to a third party subject to certain conditions. For instance, the land use rights granted for commercial and industrial purposes is for 40-50 years and for residential purposes 70 years (See [Elosua et al. \(2013\)](#)).

The local government has discretion on land sales and has the authority to change the farming land to residential or commercial use for urbanization purposes. Nevertheless, it is the central government that controls the land supply; the central government sets target for the local government concerning the amount of land they can sell (urbanization goal). The local government can then decide whether to develop the land for residential or commercial use and at what price. Also, the Central Economic Working Conference, organized every December jointly by the Central Committee of the Communist Party of China and the State Council, summarizes the economic performance of the given year and sets out the national GDP growth target for the subsequent year. From the national GDP growth target, the 31 provinces in China have to specify their GDP growth target. This mechanism creates a perverse incentive for local government officials to achieve their respective targets at all cost, as their performance is analyzed based on the achievement of their growth target ([Fan 2015](#)).

The 1994 reform of the tax system termed as the “tax sharing system” provided central government more autonomy over tax revenue. The central government increased its share in national income by disproportionately increasing the share of local government contribution in central tax revenue. Incidentally, at the same time, the central government passed a budget law that forbids local governments from running budget deficits and raising funds by selling bonds or through bank loans. These measures have therefore forced local governments to create unique structures to finance their expenditures; these structures are called Local Government Financing Vehicles (LGFV’s). The local governments provide capital to these financing vehicles through cash, transfer of land use rights or infrastructure assets namely roads and bridges. Once the financing vehicles have met the capital requirement, they then raise funding through bank lending or equity and bond markets. The 2008 stimulus further accelerated the creation of these financing vehicles and by mid-2009 more than 3,800 LGFV’s were in China which by 2013 were more than 10,000¹⁰. Figure 7 shows, that a significant part of the underlying capital of these financing vehicles comprised of land usage rights. As we have discussed previously, the 2008 stimulus increased land prices tremendously and henceforth the contribution of land usage rights in local government revenue which in 2010 peaked at more than 35% of the central government revenue. However there are some risks associated with this, arising from implicit and explicit local government guarantees (local governments have assurances from the central government) for these financing vehicles, thereby creating perverse incentives for banks to lend to these vehicles. Also, during an economic boom period as land prices start increasing the financing vehicles will try to extract more loans for a given amount of land, however in times of an economic slowdown the value of the collateral will fall, in such circumstances the banks will demand security to manage their risks. If the financing vehicles are not able to provide sufficient security, then the banks will be forced to sell the underlying collateral. However these asset sales in an economic slowdown can put downward pressure on asset prices, reducing the value of the collateral, and thereby creating a financial crisis.

¹⁰BBVA 2011 Report: Who will pay the bill for local governments’ fiscal stimulus?

We look at the Chinese Residential Land Price Index developed by [Wu et al. \(2012\)](#) at the Wharton/NUS/Tsinghua universities. This index tracks the change in Chinese residential land prices since 2004 on a quarterly basis, according to this index as we see in figure 8, the residential land prices have risen by five times in the past ten years. However on close examination, we observe that real estate prices having bottomed out by the end of 2008, have since more than doubled by 2011. Thereby reflecting the effect of the stimulus and ever since rising by another 50 percent until the end of 2014. We further look into the regional level data figure 9, which is on a half yearly frequency, the data shows that prices after reaching their peak in 2013 have started falling in the western region which comprises of Xian, Chongqing, Chengdu cities and since 2014 in the middle region. If one further looks at the city level data figure 10, we see a similar picture wherein the 12 major cities in China excluding Beijing, the land prices have started falling after reaching their peak in the second half of 2013. However in Beijing land prices are still rising and have risen more than 11 times since 2004, and have more than doubled between 2012 and 2014.

2.3. Foreign Exchange Market

Money serves as a store of value and medium of exchange; it is the lubricant that enables a society to organize itself to achieve economic progress. The ability to store the fruits of one's labor for future consumption is necessary for the accumulation of capital, the spread of technological advances and, as a consequence, rising standards of living. Before 1994, China operated a highly divergent and distinct dual exchange rate policy, wherein foreign investors were not allowed to own renminbi. In the dual exchange rate regime, there was an overvalued official exchange rate and a market determined by domestic and foreign businesses trading in currency swap centers. Therefore, International visitors were obliged to exchange their hard currency at the official rate into Foreign Exchange Certificates, which can then be used in Mainland China to buy goods and services. Starting from 1994 as China's economy opened up to global trade, private businesses flourished, the dual exchange begun to break down. In January 1994, China

abolished the dual exchange rate system and unified the exchange rate at the same time devaluing the Yuan 33 percent overnight as part of the reforms. China further allowed the Yuan to be fully convertible on the current account by 1996 (See [Prasad et al. \(2005\)](#)).

The Chinese Yuan strengthened from 8.7 Yuan against the US Dollar to 8.28. However, the Chinese central bank managed to keep the Yuan between the currency band of 8.2770 - 8.2800 during the Asian crisis by frequently intervening in the foreign exchange markets. In 2001, China joined WTO and confirmed its willingness to undertake gradually more financial sector reforms. The landmark moment came in July 2005 when China decided to exit the Yuan fixed exchange rate arrangement with the US Dollar since 1996. The Chinese authorities decided to go for a managed floating exchange rate (Yuan will be allowed to trade within the daily trading band set at $\pm 0.3\%$). Thereby the exchange rate will be increasingly determined by market forces referenced to a basket of currencies including US Dollar, Euro, Japanese Yen, Korean Won being the main currencies in the basket, as these are the currencies of China's major trading partners in the global trade. In 2007 just before the onset of the global financial crisis, the trading band was widened to $\pm 0.5\%$. However in July 2008 as the Global economy was slowing down the Chinese authorities entered into a fixed exchange rate system wherein the Yuan was fixed at 6.83 against the US Dollar to provide a boost to the export sector. In 2010, China embarked on a pilot project as part of the financial reforms to internationalize Yuan, The Chinese central bank allowed Chinese and Foreign companies to settle trade transactions in Yuan. The Chinese central bank undertook economic reforms and let the trading band for Yuan against the US Dollar to widen from $\pm 1\%$ in 2012 to $\pm 2\%$ by 2014. To maintain its credibility in the maintenance of the currency arrangement the Chinese central bank has amassed close to US Dollar 4 Trillion of reserves by 2014¹¹.

The general price level that is, the average exchange rate for money against all goods and services, and how it changes over time, plays a profoundly important role in any

¹¹PBOC

society because it influences the nature and scope of our economic and social relationships over time. For instance, in China as the real GDP per capital has risen faster than its trading partners, therefore, the renminbi should rise relatively to other currencies when taking into account wage and productivity differentials to clear any imbalances. Since the renminbi is pegged to the US dollar, the Chinese central bank to offset the demand for Renminbi and maintain the currency peg has amassed vast claims on foreign assets. The Chinese central bank has been forced to undertake massive monetary easing operation by selling claims on domestic assets in the form of Renminbi, for instance, broad money supply (M2) in the economy has grown faster than the growth in nominal GDP¹². The quasi-fixed exchange rate system that the Chinese monetary authorities have maintained has indeed provided stability to Chinese economy against business cycle fluctuations. Further, with a quasi-fixed ERM with the US Dollar, the Chinese Monetary authorities are importing US Federal Reserve monetary policy. This type of an arrangement is beneficial when relative to other currencies the US Dollar is cheap, as the Chinese exporters benefit from a weaker currency. However as the US Dollar starts appreciating relatively against other currencies, Chinese exporters suffer due to an expensive currency. For instance, as we see in figure 11, the Renminbi has risen in real terms against its trading partners from 2006 till 2015, by more than 50% thereby eroding Chinese export firms competitiveness in the global markets, whereas at the same time increasing the real purchasing power of renminbi denominated assets. The Renminbi real exchange rate movement exhibits similar movements to the Japanese Yen between 1985-94, before the asset bubble in Japan went bust in the 1990's. This arrangement will create challenges for policymakers as a stronger US Dollar may not be highly desirable for the local economy, as it may engender asset bubble. Lastly, if individuals holding Yuan denominated assets foresee in future that the value of Yuan will depreciate relative to the US Dollar, they may most certainly be willing to exchange Yuan denominated assets to other foreign assets, these self-fulfilling expectations coupled with significant capital outflows can lead to a financial crisis (See Obstfeld (1996)). Therefore, in these conditions, the present mechanism will eventually

¹²The Peoples Bank of China & National Bureau of Statistics

make it costly for the central bank to maintain the currency peg and ensure financial stability at the same time.

3. Shadow Banking System

The role of shadow banks in emerging economies has been widely misunderstood, for instance, entities such as microcredit companies, money lenders, and pawn shops, form an integral part of the financial system. These non-banking entities integrate finance with the real economy, notably in providing access to credit to marginalized sectors which may not qualify for loans from the formal banking industry due to onerous regulations, lending requirements, lack of credit information and gaps and overlaps in regulatory coverage. Shadow banks also fulfill market needs for longer-term financing which the official banking sector may not meet, as it is biased towards short-term lending as longer-term loans attract higher capital and other regulatory requirements. It is important to note that not all “shadow banking” activities are toxic. Many (such as pawn-broking, money lending and Microfinance) perform a valuable function in providing access to credit to market segments that are underserved by the traditional banking system. Historically, non-bank financial intermediaries have always been an important and integral part of the financial system, in both the advanced and developing economies (See [Acharya et al. \(2013\)](#); [Claessens and Ratnovski \(2014\)](#); [Yeva and Randall \(2010\)](#)). However, in the aftermath of the global financial crisis shadow banking system came into limelight for its role in amplifying financial institutions leverage and risks to the financial system during the crisis, for instance [FSB \(2012\)](#) defines shadow banking as “*credit intermediation involving entities and activities (fully or partially) outside the regular banking system*”. FSB¹³ considers that the most significant risks to financial stability are likely to materialize through the non-bank credit intermediation channel. Thereby suggesting that regulators should focus on credit intermediation outside the regular banking system that raises *Systemic Risk* concerns, by means of maturity/liquidity transformation, leverage and flawed credit risk transfer and/or through regulatory arbitrage concerns (See [Cetorelli and Peristiani](#)

¹³Financial Stability Board

(2012); Shin (2009); Adrian and Ashcraft (2012)). Pozsar et al. (2013) define shadow banks as “financial intermediaries that conduct maturity, credit and liquidity transformation without direct access to central bank liquidity or public sector credit guarantee”. However this definition seems to neglect that many shadow banking activities operate within banks importantly SIFI’s, ¹⁴ we, therefore, consider shadow banking as “ALL FINANCIAL ACTIVITIES, EXCEPT TRADITIONAL BANKING, WHICH REQUIRE A PRIVATE OR PUBLIC BACKSTOP TO OPERATE” see Claessens and Ratnovski (2014).

Before we delve into the shadow banking system in China, we discuss the build-up of private debt in the Chinese economy, for instance, credit to the private non-financial sector¹⁵ stood at 193% of the GDP in 2014 up from 120% in 2006 (See figure 12). However, more than 85% of the debt is corporate debt. The data in figure 13 further highlights that even though the GDP has risen by three times in 2014 from its 2003 level, overall credit has grown by more than eight times for the corresponding period. This further explains the role of debt in the growth story more notably after the financial crisis. However before moving it is important to consider the Chinese banking system in light of the non-market policy instruments that PBOC uses to conduct monetary policy. For instance Loan to Deposit Ratio (LDR) are used as a policy tool to regulate the amount of credit in the economy. Since 1998 the Chinese central bank publishes aggregate loan quotas for the banking industry on a annual basis. LDR’s are stipulated by PBOC which at present stands at 75% meaning Chinese banks are allowed to lend out only 75% of their deposits as loans. Prior to 1998 PBOC had predetermined loan quotas for commercial banks, the loan quotas were announced at the beginning of the year and remain constant; the purpose of this policy tool has been to reign aggressive lending. This policy tool is rather unusual in modern financial markets as central banks often use short-term interest rates coupled with Reserve Requirement Ratio (RRR) to steer the economy based on their mandate. For instance figure 14 shows the widening gap between bank loans and total

¹⁴Systemically Important Financial Institution

¹⁵This is excluding Government Debt which further accounts for 55% of GDP

credit creation as a percentage of GDP, this difference captures credit creation outside the formal banking system, and, we term this gap as the “Alternative Financing”. We further compute credit gap in figure 15, which is the deviation of current credit growth from its long-term trend¹⁶. This measure is a useful early warning indicator (EWI) for banking crisis (see [Borio and Lowe \(2002\)](#); [Drehmann and Tsatsaronis \(2014\)](#)). Banking crisis tend to arise primarily from deteriorating economic fundamentals, most notably declines in asset quality due to exposure of different institutions to common risk factors. This further reflects vulnerabilities that have manifested over an extended period, by the mutual interaction between the financial sector and the real economy. For instance, as the economy expands, asset prices increase, risk premia tend to decline and, as a result, financing becomes cheaper and more plentiful. These developments lead to a rise in real incomes through productivity growth. However if this expansion goes unabated and investments are not based on a risk return metric, but on ever increasing asset prices and extraction of equity from real assets. It is this sense of euphoria that will eventually sow the seeds of subsequent contraction, wherein unproductive investments will be written off, and losses will accrue to equity holders. Unless the financial system has built up sufficient capital buffer¹⁷ during the boom time, the subsequent contraction can lead to widespread instability. Ex-post, a financial cycle is evident. We further look at the total social financing/total credit data reported by the People’s Bank of China (PBOC) in figure 16. We observe a significant jump in credit creation by Entrusted loans which accounted for less than 4% of total bank credit in 2002 and by 2014 credit provided by entrusted loans rose to 11% of total bank lending and in GDP terms accounted for more than 14.50% of GDP.

3.1. *Entrusted Loans*

Entrusted loans are loans made by non-financial firms/individuals to other designated non-financial firms via the traditional banking system. For instance, ICBC the biggest

¹⁶The credit gap is calculated using a one-sided Hodrick-Prescott filter with a smoothing parameter of 400,000.

¹⁷equity capital by definition is the buffer between solvency and bankruptcy.

Chinese commercial bank defines entrusted loan as a loan where ICBC acts as an agent of the funds from the "Principal" (government departments, enterprises/public institutions or individuals) and administers the funds according to target borrowers, purpose, amount, term, and rate. The bank only collects the handling charges and will not undertake any loan risk. Therefore, the banks act as a trustee and, in particular, the bank does not assume any credit risk and does not have to set aside any funds to satisfy capital requirement rules.

Entrusted loans, in general, have been used by both government and private entities for Intragroup cash management. For instance, local governments often provide entrusted loans to support Small and Medium Enterprises SME's and other industry sectors. Intragroup cash management often associated with large companies wherein they use entrusted loans to allocate funds among subsidiaries. In most cases Company A has two subsidiaries X and Y. Subsidiary X has excess reserves sitting in the bank earning deposit rate while subsidiary Y requires credit from the bank which is at a higher rate than the deposit rate. In general, the treasury function of the Company A will ensure the flow of funds from subsidiary X to Y, however in China, this lending will be considered as private credit as there is no financial institution involved and is illegal. Therefore such intragroup fund allocation has to be done via entrusted loans. Another category of entrusted loans is of interest rate arbitrage to curb private lending regulators encourage entrusted loans thereby creating more transparency and serving the needs of the private sector and SMEs. This means that companies having access credit or easy access to cheap bank credit can grant entrusted loans at relatively high-interest rates to credit-starved borrowers and profit from such interest rate arbitrage. In the final category, the entrusted loans are packaged into Wealth Management Products (WMP's)¹⁸. Since 2013, a new driver for entrusted loans has been the exponential jump in wealth management products packaged by brokers or Targeted Asset Management Companies (TAMC), similar to trust loan products. As fig-

¹⁸Wealth Management Products (WMP's) are short term investments that typically offer fixed rates of return to private investors with a minimum investment of 50,000 yuan. These products are offered by Asset Management Companies and are sold by banks.

ure 17 shows in this category, the banks will be involved either as an originator and/or distributors of such products.

In the absence of data the actual usage of entrusted loans is hard to analyze. However based on industry research and two surveys carried out by CBRC, close to 45% of Entrusted loans fall in the interest rate arbitrage and WMP category¹⁹. The other side of this picture also shows that 45% of the entrusted borrowers, cannot avail credit from the formal banking sector, or are considered risky by banks. It is in line with the broader economic picture as we see negative growth in land prices and a slowdown in industrial production coupled with excess capacity and increase in Total Social Financing. As SOE's and large enterprises have access to credit, entrusted loans have become a new source of income for these enterprises, as they can take out cheap bank credit and lend it at higher interest rates to other firms in the economy. The real estate industry on the other hand already facing an economic slowdown coupled with a credit crunch due to restrictions on credit availability by the Chinese regulators, finds entrusted loans as a new source of albeit expensive funding a good funding option in order to avoid liquidation at fire sale prices and bankruptcy. However this unsecured financing arrangement raises issues concerning the financial stability and resilience of the financial system. Firstly the business sector presumably does not possess the technical expertise required for credit appraisal, as it is the business entity which is undertaking the credit risk, and needs to appropriately price risk. Secondly, the business entity through interest arbitrage and WMP loans are providing credit to borrowers who do not have access to formal credit offered by financial institutions; they are in essence giving credit to borrowers deemed unworthy by the banks in the first place based on a risk return metric. It's a critical mechanism as the business entity is borrowing money from banks through the formal channel and lending them back to borrowers deemed unworthy of credit by the banks themselves. Therefore in essence even though all the credit risk is assumed by the business entity taking out the loan, but indirectly the bank is exposed to higher risk because of this high-risk lending.

¹⁹Wang Jiahui CBRC “Wei Tuo Dai Kai Ye Wu de Te Zhe” (2013), LI Mei & SUN Yan-na (2013)

In the case of a default by the entrusted borrower, the lender will also default, and the probability that the lender may default on the bank loans is presumably quite high in comparison to the risk weighting applied by the bank loan officers at the beginning of the loan appraisal process. Lastly, the Chinese banking system is an arms length based financial intermediation system, as most of the credit creation happen through banks. In this kind of a system as banks have sophisticated and detailed information about both the financial health of borrowers as well as lenders having excess cash or access to credit, the banks can utilize this information asymmetry. For instance banks have information concerning borrowers having difficulty in repaying their loans. The bank, in this case, can act as a conduit whereby it substitutes troubled loans in its portfolio, by entrusted loans offered by big firms having excess cash, thereby shifting credit risk from its books onto the big company’s balance sheet via entrusted loans. Additionally, if the banks discover that a borrower is having difficulty in making repayments the bank can provide funds through the interbank market, by using off-balance-sheet investment vehicles like Trust Asset Management Companies. Through this mechanism, the bank doesn’t have to set aside any capital as interbank lending is considered risk free and attracts zero risk weights. This frees up bank balance sheet and allows them to provide funds to borrowers that they cannot offer through the formal banking channel.

To further understand the impact of entrusted loans on bank balance sheets during different stress scenarios, we collected the loan data on 100 major Chinese financial institutions for 2014, and our data covers 80% of the total Entrusted Loan origination in China for the fiscal year of 2014. We further restrict our analysis to 15 primary entrusted loan originators in China. As these 15 institutions account for more than 70% of the Entrusted Loan origination. Our sample also covers six Systemically Important Financial Institutions (SIFI’s) ²⁰ in China, as defined by China Banking Regulatory Commission (CBRC) for banks having assets of at least 1.6 Trillion Yuan. Our dataset further com-

²⁰China Construction Bank, Agricultural Bank of China, Bank of Communications, China Merchants Bank Corporation, Bank of China & Industrial and Commercial Bank of China

prises of four Global SIFI's as designated by the FSB in 2015 namely Agricultural Bank of China (ABC), Bank of China (BOC), China Construction Bank (CCB) and Industrial & Commercial Bank of China (ICBC). As we have discussed before banks act as a trustee in Entrusted loans, and as the bank is not assuming any credit risk, there are no risk-weighting attached to these loans by loan officers, and consequentially banks are not required by IFRS to report corresponding data on these loans. In our sample of the 15 banks Agricultural Bank of China, Bank of China and Bank of Communications are not reporting the entrusted loan data since 2007-08. We estimate the share of these three financial institutions by calculating the proportion of entrusted loans as a percentage of the total entrusted loans last reported by these banks and consider the same ratio for 2014. As we report in Table 1, according to our estimates Agricultural Bank of China has 9.5% share of total entrusted loans, Bank of China 7.6% and Bank of Communications 6.10% respectively. Our estimates seem quite reasonable as China Construction Bank and ICBC have a share of 14% and 8.4% respectively, and when looking at the percentage of Entrusted Loans to Gross Loans issued in 2014, the estimates are quite well in line.

3.2. Methodology

To analyze the impact of an idiosyncratic economic shock and its effect on the balance sheet and profitability of these financial institutions, we run stress tests on the 15 banks in our database. We use Cihak (2007) methodology for our stress test as we are interested in examining the effects of an idiosyncratic shock to the economy, and the impact of the shock on Chinese banking sector. In our stress test scenario, we look at the effect of the shock both on the banks' balance sheet and their profitability. Our stress tests consider the case if banks have to take entrusted loans on their balance sheets and are required to make capital provisions against these loans and attach risk weights. As entrusted loans are off-balance sheet items and in the absence of information concerning the usage of these loans, it's hard to ascertain the actual beneficiaries of these funds. However based on survey data on 10 Chinese banks, around 55% of the entrusted loans have been used by firms for LGFV loans/intra-company transfers and rest for interest rate arbitrage and lending through Trust Asset Management Companies. We are interested in this category

which presently covers 45% of the Entrusted Loans. As there is no data available on the usage of entrusted funds, we rely on survey data, which suggests that 25% of entrusted loans have gone to the Real Estate sector which has been experiencing a slowdown since 2013.

Tables 2 & 3 document the two scenarios for our stress tests wherein banks have to bring 25% and 45% of their entrusted loans on their balance sheet, presumably due to regulatory requirements or due to a “Liquidity Crunch” in the wholesale interbank funding market. We further consider 100% risk weighting on these loans, however some portion of the entrusted loans are backed by bank loans and are assigned Risk Weights, hence provisioned for losses. Therefore, in our stress test scenario, we consider 25% of the Entrusted Loans for all the banks been assigned 100% Risk-Weights on the banks balance sheet and of the remaining 20% have been assigned 50% Risk-Weights. At individual bank level out of the systemically important banks, Bank of Communications will undergo more than 4% Risk Weighted Asset Inflation. Our results also show that Bank of Shanghai, Bank of Jiangsu & Hua Xia Bank will experience more than 5% Risk Weighted Asset Inflation, in particular, Hua Xia Bank experiencing close to 6.5% increase in its Risk Weighted Assets. However when looking at the effect on Tier-1 Capital Ratios in both the scenarios, of the domestic SIFI’s only Bank of Communications suffer close to a percentage point hit in the stress test, even after the test its Tier-1 ratio remains at more than 10.50. Even after the stress test scenarios the SIFI’s Tier-1 capital ratios remain well above ten percentage points. Our results further provide evidence that of all the banks in our dataset, seven banks can have their Tier-1 ratio between 7-8%, with the only exception of Hua Xia Bank, which can see its Tier-1 ratio to fall below 8% in both the stress test scenarios.

In Table 4, we consider the scenario wherein banks have to write down 5% of their reported entrusted loans, and its effect on their profitability. At individual bank level, we see the SIFI’s will not be much affected apart from Bank of Communications which will see close to 30% hit to its Pre-Provision Profit. However apart from Bank of Communica-

tions, seven other banks will have their Pre-Provision Profit scored between 30-40%, with the notable exception of Hua Xia Bank, which will see its Pre-provision Profit affected by more than 45% close to its half-yearly earnings. As we see from all the banks in our database, Hua Xia is the one most exposed to Entrusted loans as a share of Gross loans at 33% second only to Industrial Bank, which stands at 36%. However when looking at the Pre-Provision Profit as a percentage of RWA, Industrial Bank is far more efficient relative to Hua Xia Bank and has high capital buffers to absorb economic shocks. When one looks at the big 6 SIFI's the banks are sufficiently capitalized and have adequate capital buffers. To fully write off the NPL's of the stress test they may have to give up two-quarters of their earnings, but their balance sheet will remain largely intact. However if the rapid increase in Entrusted Loans as have been observed since the 2008 crisis remains at 30-40%, then the NPL ratio of the banking sector can exceed more than 10% in the 3-4 years, and can present a systemic banking crisis. To look at the magnitude in GDP terms, if Chinese banks have to write down 5% of their Entrusted Loans, the loss Pre-Provision Profit can amount to 0.9% of Chinese GDP. Alternatively, we also look at RWA density which is a balance sheet measure, calculated as a ratio of RWA to Total Assets. The lower this ratio is, the less capital a bank is required to hold against a portfolio of assets of a given value. Most of the Chinese banks RWA density has been more than 60 percentage points and in particular, both China CITIC Bank and China Minsheng Banking Corp. have Density measure of more than 70 percentage points. This measure reflects the extent of off-balance sheet activities a given financial institution has undertaken, without assigning any Risk Weights. Further this test allows us to capture the scope of off-balance sheet activities that Agricultural Bank of China, Bank of China and Bank of Communications are carrying on their books without providing proper disclosure. So we can see the other three SIFI's ICBC, China Construction Bank and China Merchants Banking Corporation have similar RWA Density between 60-61 percentage points.

People's Bank of China (PBOC), the Chinese central bank, runs its own stress tests for the banking industry. However, it does not disclose any information concerning the

methodology and the data. The IMF in its 2015 Article 4 consultation has highlighted rising credit levels and real estate sector as potential risks to the Chinese economy in the near term. Alternatively, McKinsey Institute estimated that in 2014 the total debt²¹ to GDP ratio of Chinese economy stands at 282%, of which close to 45% has gone to the real estate industry. For instance, these numbers are identical and are even significantly larger when compared with Japan and Asian Tiger economies (see Table 5) before these economies experienced their asset bubbles. Our stress test results have shown the impact of an idiosyncratic shock on the Chinese economy, and its effect on bank balance sheets through the shadow banking system. Even though the SIFIs will remain stable and capitalized adequately in our stress test results, the banks can experience contagion through the wholesale interbank market, due to banks exposure to risks created by the rise of off balance sheet interlinkages, especially to the informal financial sector. In 2014, CBRC Chairman Shang Fulin expressed concerns that Chinese banks must specifically monitor credit risks in the property sector, local government financial vehicles and industries that are suffering from overcapacity problems. As in the past, the Chinese banking system had experienced twice banking crisis, once in late 1990's when the Chinese government injected 270 Billion Yuan of capital in the big four state-owned commercial banks and transferred 1.4 Trillion Yuan of Non-Performing Loans from the banks balance sheets, and again in 2003 when the government had to inject more than 450 Billion Yuan to recapitalize Bank of China, China Construction Bank, and ICBC.

4. Conclusions

In this paper, we have studied the China's economic growth model most notably post-2008 period to foresee the effect of stimulus that the Chinese government provided to the economy in the wake of the global financial crisis. We look into China's capital markets to understand the financial intermediation process in the Chinese economy, the evolution of the real estate sector post-2008 stimulus. We further consider the massive private corporate sector debt accumulation in the Chinese economy, and we observe that the marginal

²¹Total Debt includes Government, Non-financial Corporate, Household & Financial sector.

efficiency of capital in the Chinese economy has been decreasing over time. We further look into the bank lending sector, and we see a big surge in the credit creation by the non-banking sector. We mainly focus on Entrusted Loans, which at present is a significant component of the Total Social Financing measure of aggregate credit in the Chinese economy. We have considered the case of an idiosyncratic shock to these loans, and the resultant impact it will have on bank balance sheets and their profitability. Our results indicate that the Chinese banking system can undergo the shock, however, there remain fragilities among some banks, which can exacerbate a crisis by financial institutions linkages through the wholesale Interbank market. The exponential increase in credit creation outside the formal banking sector creates challenges for PBOC whose central mission is to maintain the stability of the Chinese financial system, and to reduce and contain systemic risks if and when required.

A more pressing issue that the Chinese economic system currently faces is the substitution of private credit with sovereign credit, wherein the natural link between the debt and equity claims is lost against a firm, and the riskiness of the assets they support. Thereby both types of securities trade principally not on the earning power of the firm's resources, but on the implicit safety net provided by the government in the form of guarantees. At present the Chinese financial system is plagued by this situation wherein SOE's and LGFV's are taking credit from financial institutions which are explicitly backed by the central government. Financial markets operate efficiently only when market participants can commit to transactions with reasonable confidence and that a rational judgment for the risk of nonpayment is made and compensated for. Risk-taking is a necessary condition for wealth creation and investment funds committed to capital projects, which do not accurately reflect consumer and business preferences should incur losses and in the end, will be liquidated. The optimum degree of risk aversion is primarily governed by rational judgments about the marketplace. Active and seasoned markets pass this test almost all of the time.

We further extend our discussion to the financial instability hypothesis [Minsky \(1992\)](#) our results indicate that the Chinese economy is perhaps entering into its own Minsky Moment wherein accumulation of debt by the non-government sector pushes an economy into a crisis. These crises are often accompanied by protracted periods of low-risk premiums and ever increasing value of asset claims, leading to speculation over asset value in the distant future using borrowed money. The increase in non-bank credit creation in the Chinese economy raises a question concerning borrowing in a Minsky world ([figure 13](#)) specifically for risky borrowers that can service their debt but needs to roll over the principal, and Ponzi borrowers who rely solely on increasing asset value to be sufficient to remain solvent. However an unforeseen increase in investor risk aversion elevates risk premia and as a consequence lowers asset values and promotes liquidation of debt that supported higher asset values. This sudden liquidation of debt in the absence of adequate collateral will likely lead to a collapse in the value of asset claims. The sudden collapse in asset prices will have an adverse effect on market liquidity, which can then create a sudden and precipitous collapse in market clearing asset prices, and an overwhelming demand for risk-free securities. In principal, the risk of these structures and investments rests with the investors, typically in a market system, lenders and investors monitor and discipline the activities, including leverage, of their counterparties to assure themselves of the financial strength of those to whom they lend. However, market discipline concerning the Chinese shadow banking system has been weak to non-existent. Because the counterparty risk in the Chinese financial system is based almost wholly on the Chinese banks government ownership rather than on the underlying soundness of the institutions, regulators can no longer rely on market discipline to contain systemic risk. When the Chinese shadow banking industry was in its infancy stage, the potential for such risk, if any, was small. Regrettably, that is no longer the case. Therefore, the central problem of China's financial system stability is its reliance on debt funded economic growth and the need to expand credit to maintain high levels of economic activity. Besides, the increase in the size and complexity of the shadow-banking sector reflects structural problems in its financial markets.

Banks	Gross Loans (GL)	Entrusted Loans (EL)	EL/GL	EL/Gross EL
China Construction Bank (G-SIFI)	9474.52	1570.36	16.57%	13.97%
Agricultural Bank of China (G-SIFI)	8098.07	1067.61	13.18%	9.50%
ICBC (G-SIFI)	11026.33	939.77	8.52%	8.36%
Bank of China (G-SIFI)	8483.27	854.09	10.06%	7.60%
Bank of Communications*	3431.73	685.52	19.97%	6.10%
Industrial Bank Co. Ltd.	1593.15	580.57	36.44%	5.17%
China CITIC Bank Corp.	2187.91	524.54	23.97%	4.67%
China Minsheng Banking Corp.	1812.67	406.10	22.40%	3.61%
Hua Xia Bank	939.99	310.85	33.06%	2.77%
China Merchants Bank Corp.*	2513.92	243.80	9.70%	2.17%
Shanghai Pudong Development Bank	2028.38	199.00	9.81%	1.77%
China Ping An Trust	728.16	186.87	25.66%	1.66%
Bank of Shanghai	484.52	156.18	32.23%	1.39%
Bank of Jiangsu	488.51	126.14	25.82%	1.12%
China Zheshang Bank	259.02	76.14	29.40%	0.68%

Table 1: Summary Data for Chinese Banks for Financial Year ending 2014

* Systemically Important Chinese Banks as identified by CBRC.
all numbers are in CNY Billions, ratios are in percent.
Source: Bankscope and authors' calculations.

Banks	RWA*	Tier 1 Capital Ratio	Pre-Provision Profit
China Construction Bank (G-SIFI)	10203.64	12.12	395.51
Agricultural Bank of China (G-SIFI)	10852.62	9.46	328.69
ICBC (G-SIFI)	12475.94	12.19	458.37
Bank of China (G-SIFI)	9934.11	11.35	303.56
Bank of Communications	4164.48	11.70	120.40
Industrial Bank Co. Ltd.	2911.13	8.89	95.20
China CITIC Bank Corp.	2941.63	8.99	86.87
China Minsheng Banking Corp.	2862.71	8.59	89.93
Hua Xia Bank	1201.10	8.49	34.05
China Merchants Bank Corp.	2893.73	10.44	115.38
Shanghai Pudong Development Bank	2868.90	9.13	94.29
China Ping An Trust	895.59	8.59	24.06
Bank of Shanghai	712.42	10.38	21.04
Bank of Jiangsu	569.67	9.82	17.83
China Zheshang Bank	386.79	8.62	12.34

Table 2: Bank Balance Sheet Data

* Risk Weighted Assets
(based on end-2014 data, all numbers are in CNY Billions)

Banks	RWA* Inflation 25% taken on B/S**	RWA* Inflation 45% taken on B/S**	Hit to Tier-1 with 25% taken on B/S**	Hit to Tier-1 with 45% taken on B/S**	Tier-1 Ratio with 25% taken on B/S**	Tier-1 Ratio with 45% taken on B/S**
China Construction Bank (G-SIFI)	3.85%	5.39%	-0.45	-0.62	11.67	11.50
Agricultural Bank of China (G-SIFI)	2.46%	3.44%	-0.23	-0.32	9.23	9.14
ICBC (G-SIFI)	1.88%	2.64%	-0.22	-0.31	11.97	11.88
Bank of China (G-SIFI)	2.15%	3.01%	-0.24	-0.33	11.11	11.02
Bank of Communications	4.12%	5.76%	-0.85	-1.02	10.85	10.68
Industrial Bank Co. Ltd.	4.99%	6.98%	-0.40	-0.56	8.49	8.33
China CITIC Bank Corp.	4.46%	6.24%	-0.38	-0.52	8.61	8.47
China Minsheng Banking Corp.	3.55%	4.97%	-0.29	-0.40	8.30	8.19
Hua Xia Bank	6.47%	9.06%	-0.51	-0.70	7.98	7.79
China Merchants Bank Corp.	2.11%	2.95%	-0.22	-0.30	10.22	10.14
Shanghai Pudong Development Bank	1.73%	2.43%	-0.15	-0.21	8.98	8.92
China Ping An Trust	4.45%	6.23%	-0.37	-0.51	8.22	8.08
Bank of Shanghai	5.48%	7.67%	-0.54	-0.74	9.84	9.64
Bank of Jiangsu	5.54%	7.75%	-0.51	-0.70	9.31	9.12
China Zheshang Bank	4.92%	6.89%	-0.41	-0.56	8.21	8.06

Table 3: Banks Stress Test Taking 25% & 45% of Entrusted Loans on Banks Balance Sheet

* Risk Weighted Assets

** Balance Sheet

Banks	Entrusted Loans	Pre- Provision Profit	5% Charge off Entrusted Loans	Percentage hit of Pre Provision Profit	RWA Density
China Construction Bank (G-SIFI)	1570.36	395.51	78.52	19.85	60.94
Agricultural Bank of China (G-SIFI)	1067.61	328.69	53.38	16.24	67.94
ICBC (G-SIFI)	939.77	458.37	46.99	10.25	60.53
Bank of China (G-SIFI)	854.09	303.56	42.70	14.07	65.14
Bank of Communications	685.52	120.40	34.28	28.47	66.44
Industrial Bank Co. Ltd.	580.57	95.20	29.03	30.49	66.07
China CITIC Bank Corp.	524.54	86.87	26.23	30.19	71.07
China Minsheng Banking Corp.	406.10	89.93	20.31	22.58	71.30
Hua Xia Bank	310.85	34.05	15.54	45.64	64.87
China Merchants Bank Corp.	243.80	115.38	12.19	10.57	61.15
Shanghai Pudong Development Bank	199.00	94.29	9.95	10.55	68.37
China Ping An Trust	186.87	24.06	9.34	38.83	55.75
Bank of Shanghai	156.18	21.04	7.81	37.11	60.00
Bank of Jiangsu	126.14	17.83	6.31	35.37	54.86
China Zheshang Bank	76.14	12.34	3.81	30.85	57.73

Table 4: Impact on Bank Profitability

Based on end-2014 data, all numbers are in CNY Billions.

	Ratio of Private Non-Financial Debt/GDP* (First Year)	Ratio of Private Non-Financial Debt/GDP** (Last Year)	Bank Lending to the Real Estate Sector*** (Percentage of Total Bank Lending)
Japan	150.8	196.5	18-20
South Korea	109.1	162.9	15-25
Indonesia	59.5	74.5	25-30
Thailand	59.1	179.9	30-40
Malaysia	97.2	172.1	30-40
China	118.9	192.7	45

Table 5: Growth and Composition of Bank Lending

Data for Japan (1984-1992), S. Korea, Indonesia, Thailand, Malaysia (1989-1997), China (2006-2014)

* First Year of the Time Period ** Last Year of the Time Period

*** Japan (1992) S. Korea, Indonesia, Thailand, Malaysia (1997) China (2014)

Source: BIS

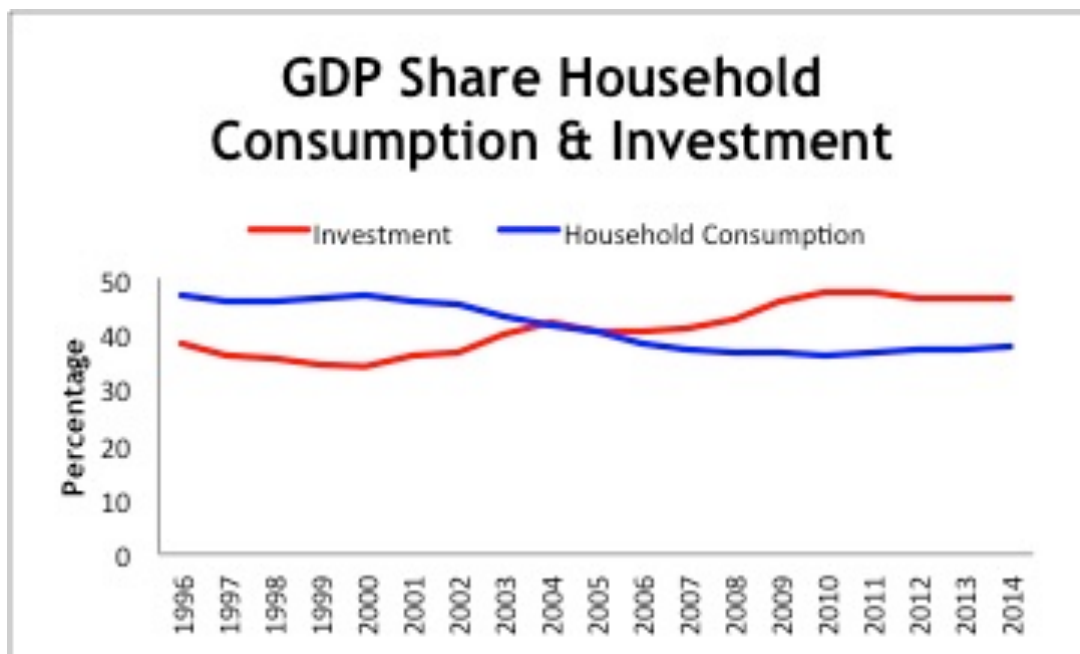


Figure 1: China's Share of Investment and Consumption % of GDP

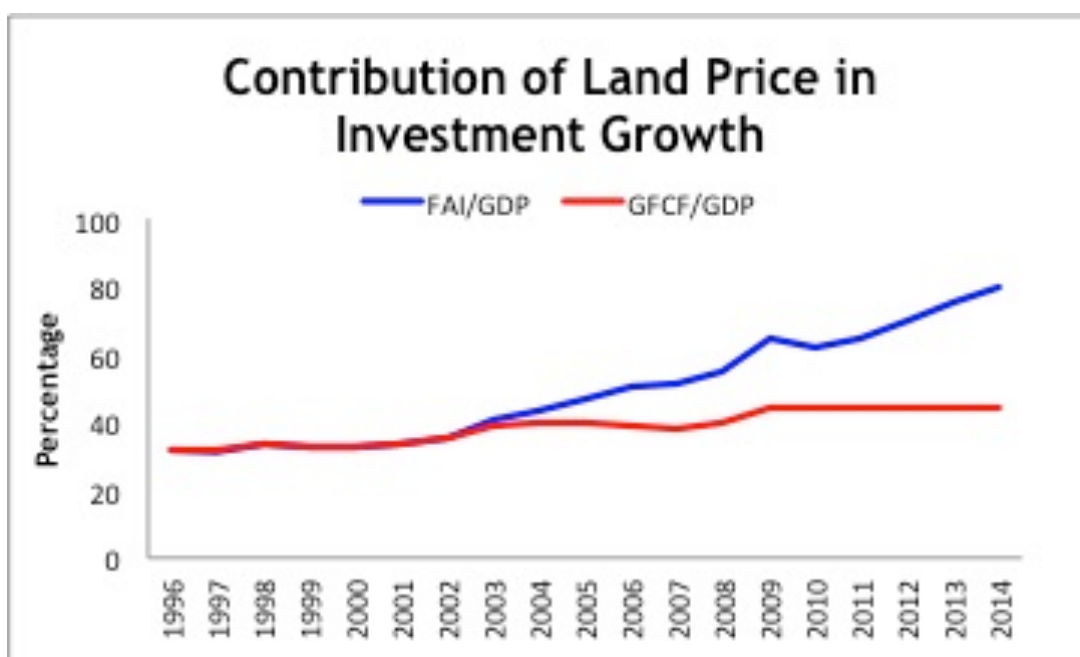


Figure 2: Land Price Appreciation in Investment Growth

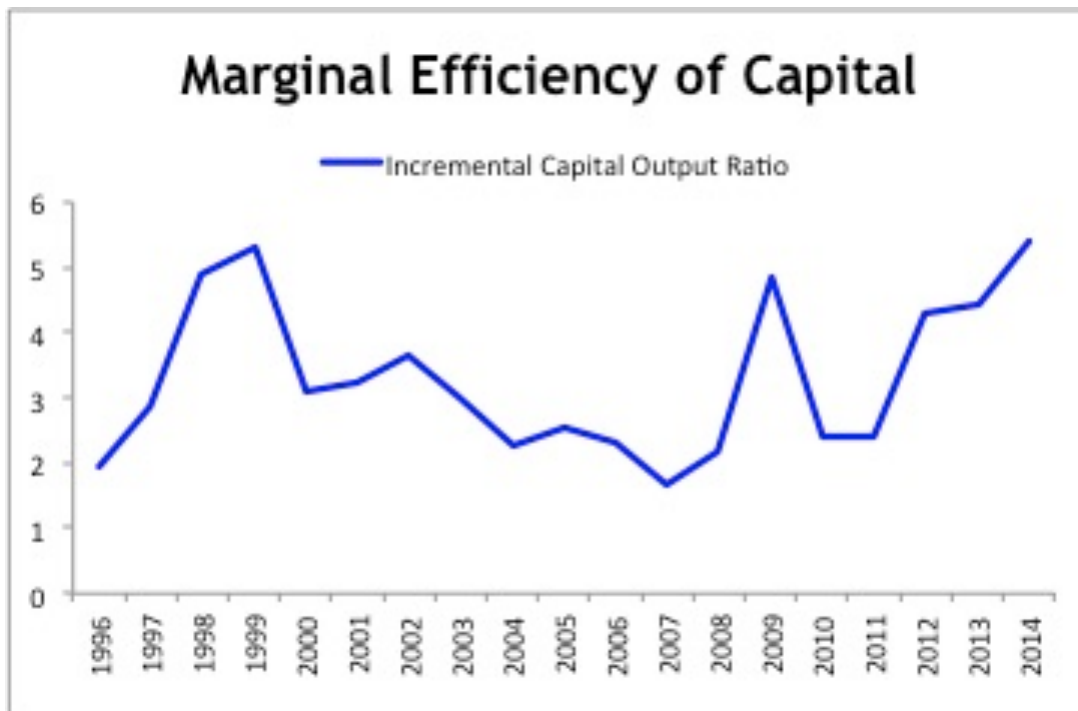


Figure 3: Marginal Efficiency of Capital

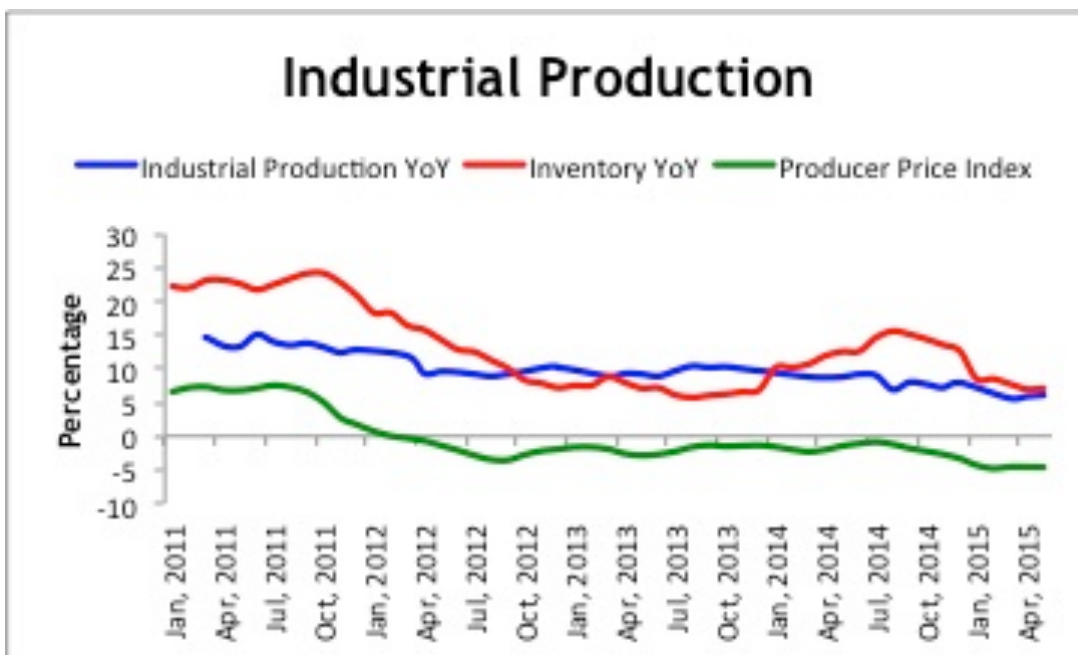


Figure 4: Economic Slowdown in Chinese Economy

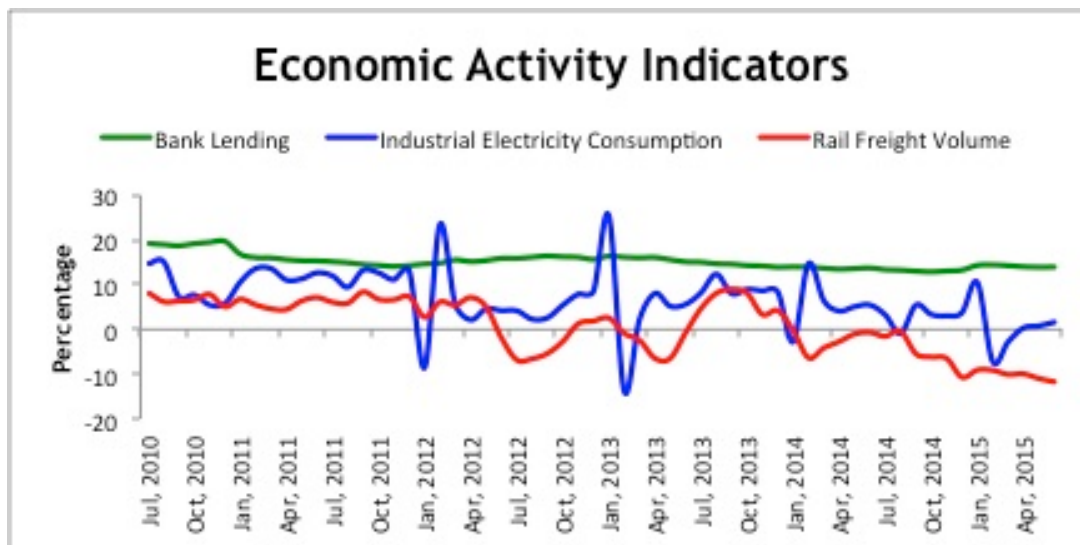


Figure 5: Alternative Economic Indicators for the Chinese Economy



Figure 6: Exports and Imports



Figure 7: Land Revenue Generating Economic Growth



Figure 8: Residential Land Price Index

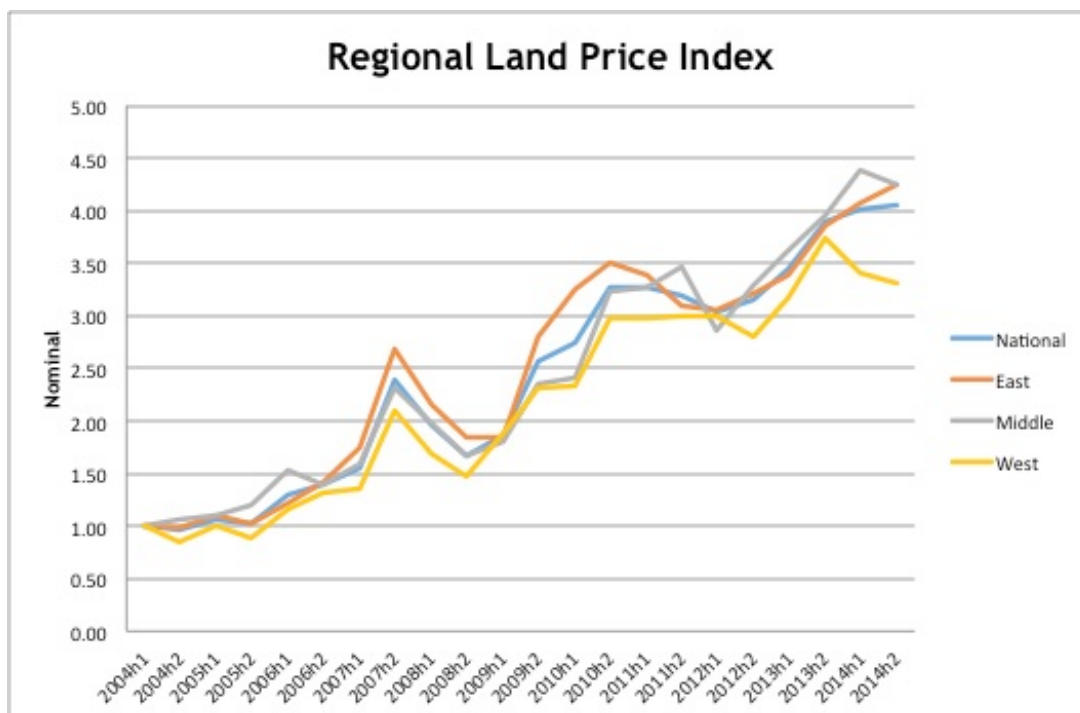


Figure 9: Regional Residential Land Price Index

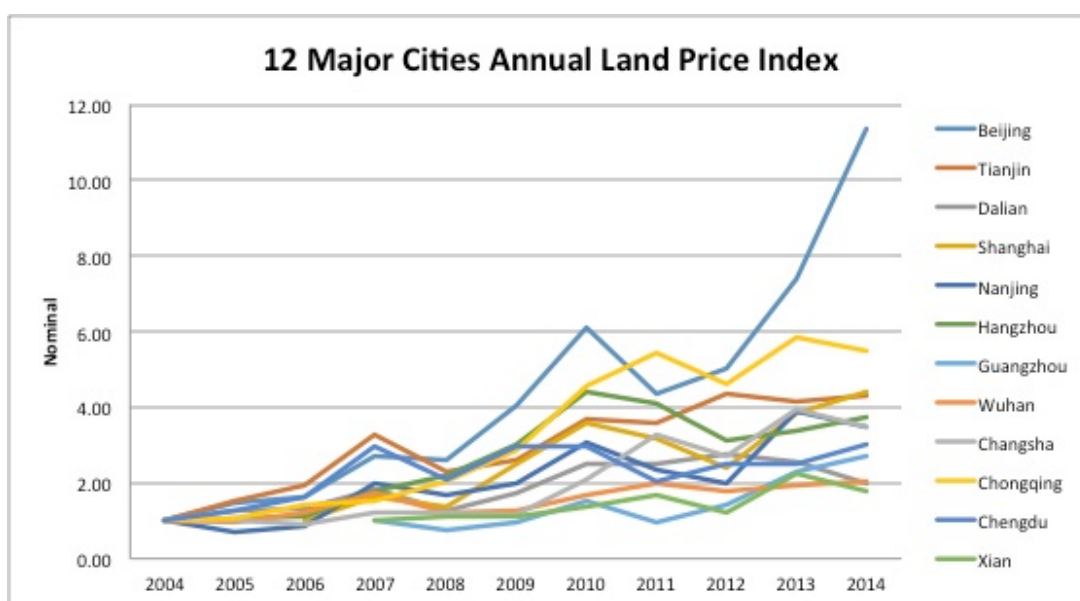


Figure 10: 12 Major Chinese Cities Residential Land Price Index

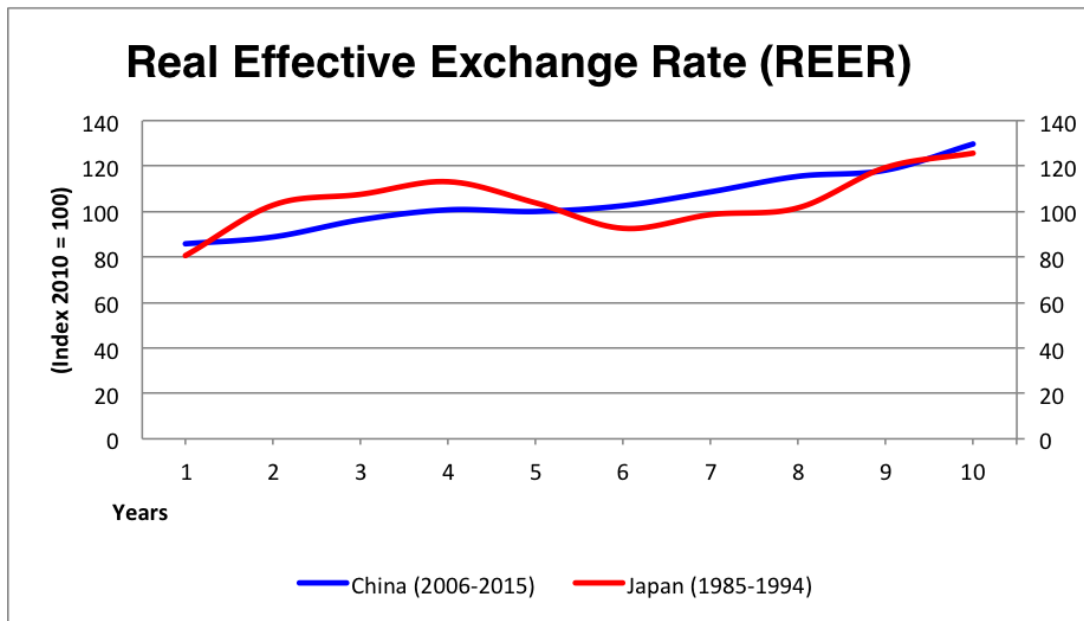


Figure 11: Real Effective Exchange Rate

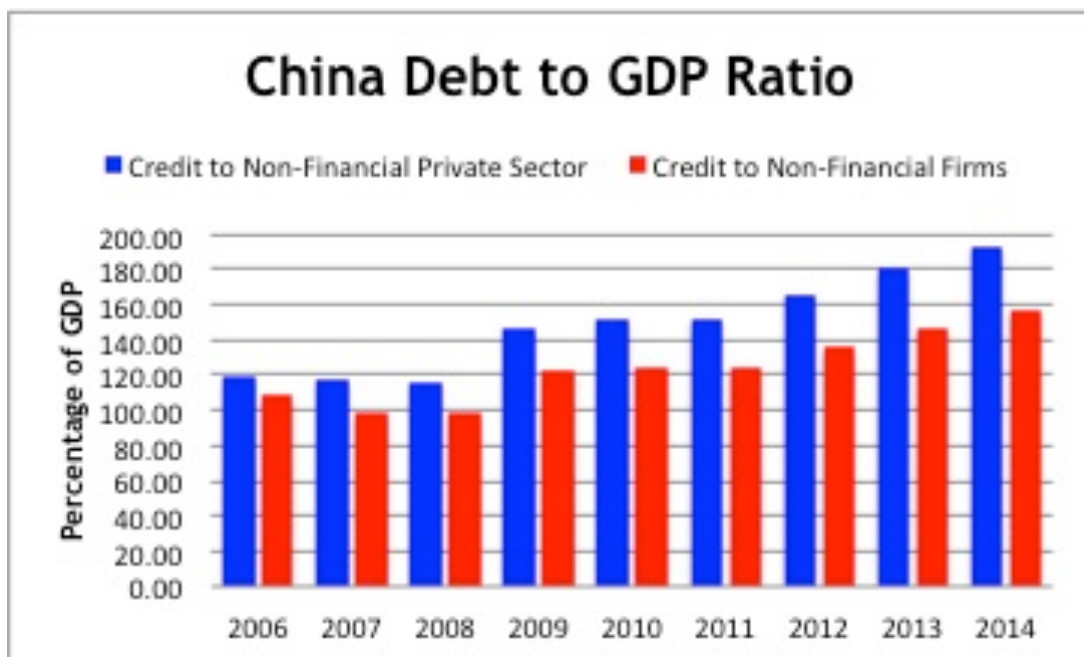


Figure 12: China's Debt to GDP Ratio (Excluding Government Debt & Financial Sector Debt)

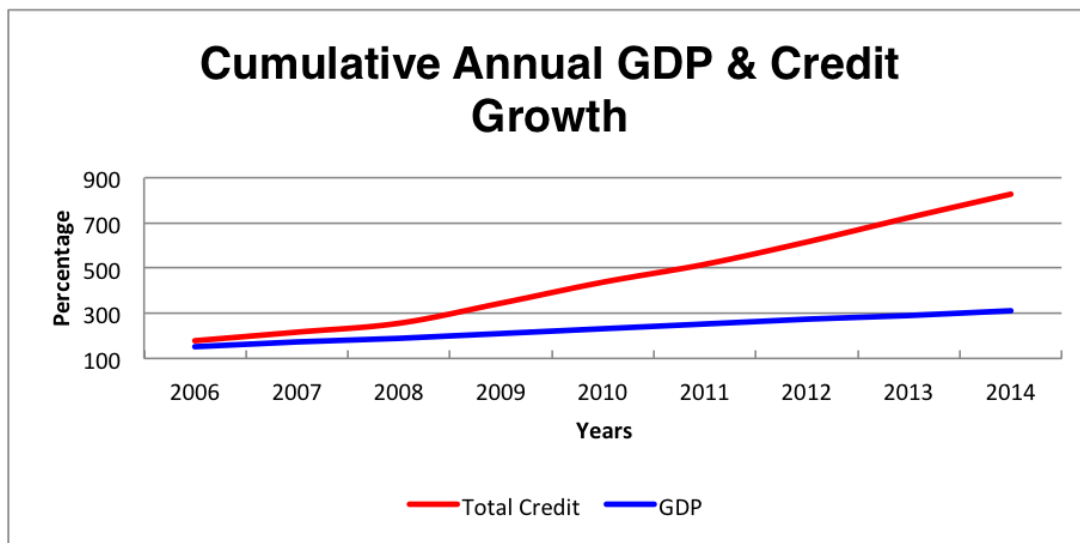


Figure 13: China's Cumulative Credit and GDP Growth

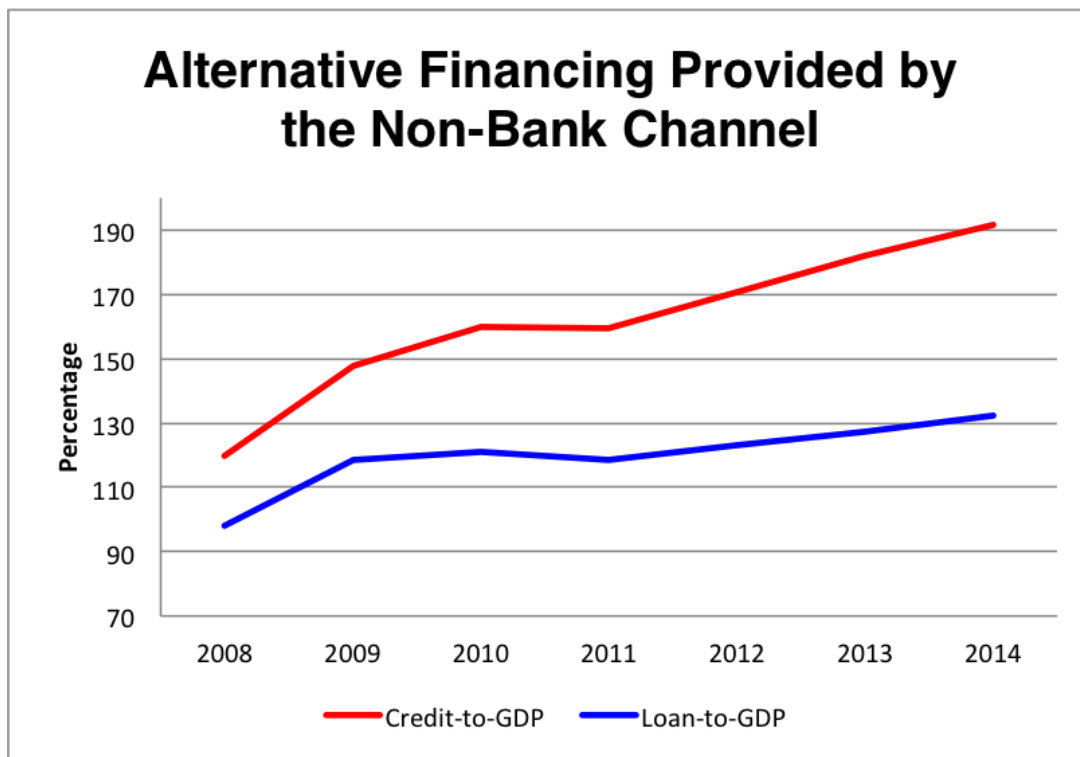


Figure 14: Credit Creation outside the Formal Banking System

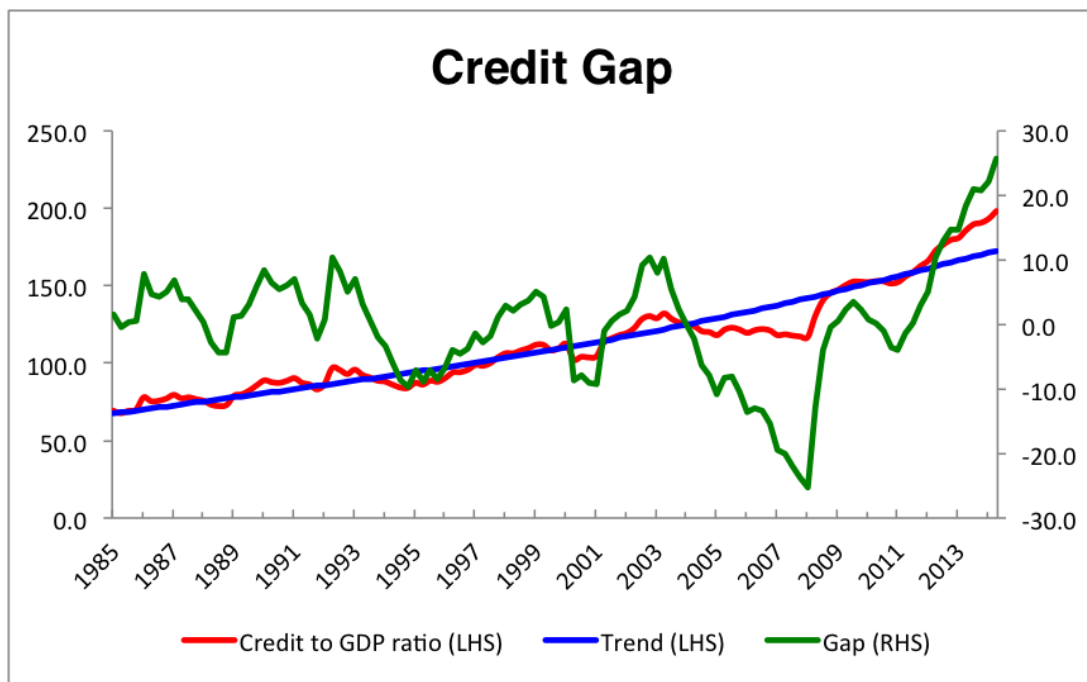


Figure 15: Credit Gap: Deviation of Credit-to-GDP Ratio from Trend

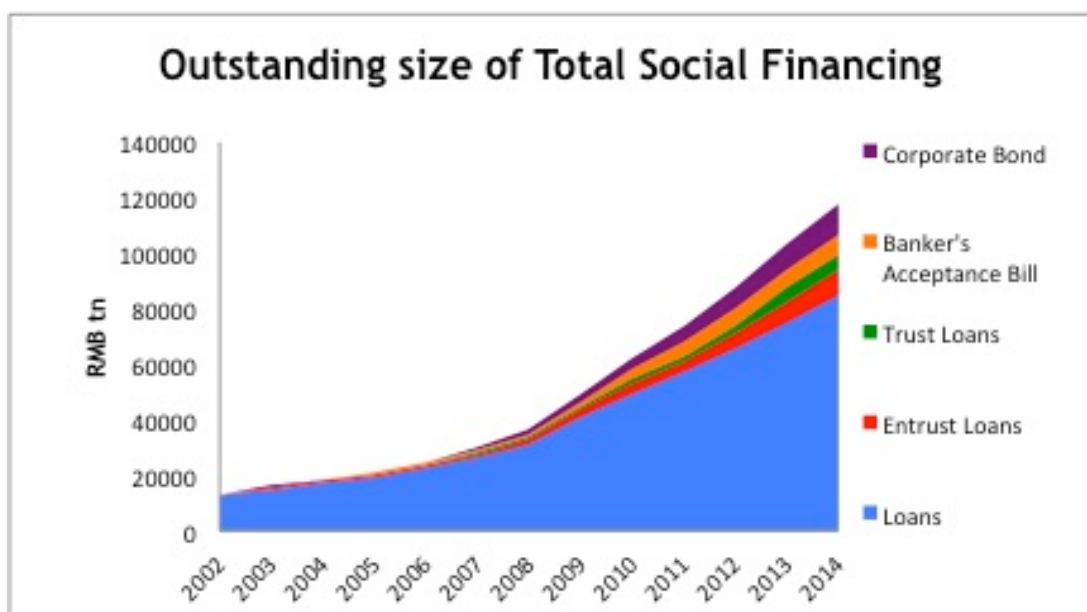


Figure 16: Total Social Financing

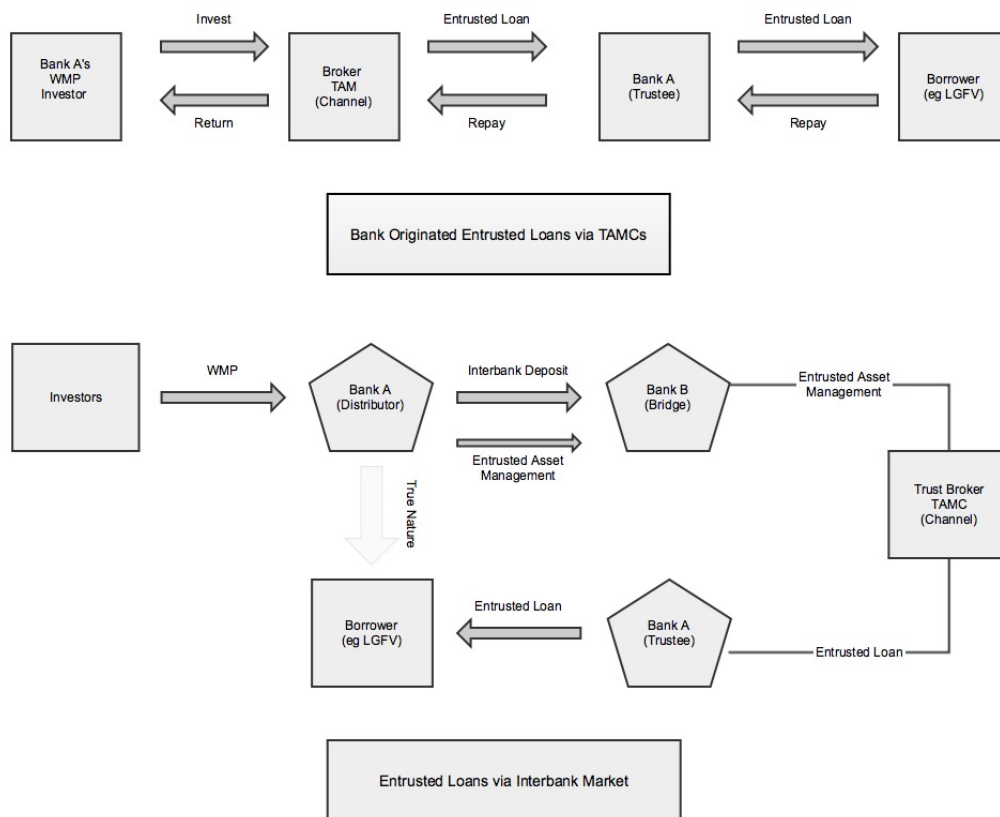


Figure 17: Entrusted Loan Structure

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